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import random

# ----- Max Heapify -----
def max_heapify(arr, n, i):
    largest = i
    left = 2 * i + 1
    right = 2 * i + 2

    if left < n and arr[left] > arr[largest]:
        largest = left

    if right < n and arr[right] > arr[largest]:
        largest = right

    if largest != i:
        arr[i], arr[largest] = arr[largest], arr[i]
        max_heapify(arr, n, largest)

# ----- Max Heap Sort -----
def max_heap_sort(arr):
    n = len(arr)

    # Build Max Heap
    for i in range(n // 2 - 1, -1, -1):
        max_heapify(arr, n, i)

    # Heap Sort
    for i in range(n - 1, 0, -1):
        arr[0], arr[i] = arr[i], arr[0]
        max_heapify(arr, i, 0)

# ----- Main Program -----
n = int(input("Enter number of elements: "))

arr = [random.randint(1, 100) for _ in range(n)]

print("\nOriginal Array:")
print(arr)

max_heap_sort(arr)

print("\nSorted Array (Ascending Order):")
print(arr)
```

Enter number of elements: 7

Original Array:

[38, 1, 57, 8, 10, 60, 74]

Sorted Array (Ascending Order):

[1, 8, 10, 38, 57, 60, 74]